Train, Validation and Test Sets

Size and churn mix of each set (Jira T7-32)

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1 How the Sets Were Made

The encoded dataset of 7,043 customers was split 70/15/15 with a stratified split on Churn (`random_state = 42`), so every set keeps the same share of churned customers. The split was done in two steps: 30 per cent was first held back, and that part was then divided equally into validation and test sets. `tenure` and `MonthlyCharges` were then standardised with a scaler fitted on the training set only, so no information from the validation or test sets was used.

2 Size and Churn Mix

Table 1
Rows and Churn Mix in Each Set

Set	File	Rows	Share	Churn = 1	Churn = 0	Churn rate
Training	`ats_train.csv`	4,930	70.0%	1,308	3,622	26.5%
Validation	`ats_val.csv`	1,056	15.0%	280	776	26.5%
Test	`ats_test.csv`	1,057	15.0%	281	776	26.6%
Total		7,043	100%	1,869	5,174	26.5%

The churn rate is almost identical across the three sets and matches the full dataset (26.5 per cent). Churners are the minority class in every set, about one customer in four, so accuracy alone can be misleading; recall, precision and F1 for the churn class should also be reported.

3 File Contents

- Each file has 13 columns: 12 numeric features and `Churn` (1 = Yes, 0 = No) as the last column.
- Features: `gender`, `SeniorCitizen`, `Dependents`, `tenure`, `PhoneService`, `MultipleLines`, `MonthlyCharges`, `InternetService_DSL`, `InternetService_Fiber_optic`, `Contract_Month_to_month`, `Contract_One_year`, `Contract_Two_year`.
- `tenure` and `MonthlyCharges` are standardised; all other features are 0/1 values.
- `scaler_params.csv` holds the training mean and standard deviation: tenure 32.3105 and 24.5833; MonthlyCharges 64.5901 and 30.1506.
- No missing values and no text columns in any file (all validation checks passed).

4 How to Use the Sets

- **Training set:** fit the ANN model.
- **Validation set:** tune settings and use for early stopping.
- **Test set:** use once, for the final evaluation only.
- Do not scale these files again; they are already scaled with the training values.
- The 302 duplicate rows were kept, so a few identical rows appear in more than one set. Test results may therefore look slightly better than they really are; this is recorded as a limitation.